

JACOBABAD, Pakistan

WMO: 417150

Lat: 28.284N

Lon: 68.450E

Elev: 56

StdP: 100.65

Time Zone: 5.00 (E05)

Period: 01-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	7.0	8.3	-3.7	2.8	19.5	-1.1	3.5	19.3	4.7	16.3	3.9	18.0	0.2	50	0.371

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
6	12.2	46.3	25.7	44.7	25.3	43.2	25.8	31.2	39.0	30.2	37.7	29.6	37.1	1.7	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
29.6	26.8	36.1	28.5	25.0	35.4	27.9	24.1	34.9	107.7	39.0	102.1	37.8	99.1	37.2	37.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 5.5	4.0	3.1	DB	4.5	49.1	2.1	0.8	3.0	49.7	1.8	50.2	0.6	50.7	-0.9	51.3	(4)
(5)			WB	2.7	33.0	2.2	2.0	1.2	34.4	-0.1	35.5	-1.3	36.6	-2.9	38.0	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	27.6	15.1	18.6	24.5	31.7	36.4	37.0	35.2	33.2	31.6	28.2	22.4	16.7	(6)	
	DBStd	7.90	2.11	2.85	3.57	3.14	2.45	1.88	1.76	1.38	1.45	2.10	2.45	2.31	(7)	
	HDD10.0	1	1	0	0	0	0	0	0	0	0	0	0	0	(8)	
	HDD18.3	193	101	29	2	0	0	0	0	0	0	0	1	61	(9)	
	CDD10.0	6422	159	239	451	649	819	811	782	719	649	564	371	208	(10)	
	CDD18.3	3573	2	35	194	399	561	561	523	461	399	306	121	11	(11)	
	CDH23.3	54354	21	301	2073	6012	9582	9773	8719	7289	5888	3588	1002	104	(12)	
	CDH26.7	36179	1	64	979	3915	7119	7377	6241	4814	3516	1824	321	9	(13)	
(14) Wind	WSAvg	0.8	0.4	0.8	0.8	1.0	1.2	1.5	1.2	1.0	0.8	0.6	0.4	0.4	(14)	
(15) Precipitation	PrecAvg	539	92	65	67	29	13	10	2	4	13	25	63	71	(15)	
	PrecMax	1278	431	251	296	147	82	55	26	38	185	252	225	263	(16)	
	PrecMin	18	0	0	0	0	0	0	0	0	0	0	0	0	(17)	
	PrecStd	305	100	74	76	36	20	15	5	8	29	40	73	68	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	25.5	30.5	39.1	45.0	48.8	48.8	44.0	39.7	38.9	37.1	33.4	28.1	(19)	
		MCWB	15.5	18.3	22.1	23.0	24.6	26.2	29.4	28.8	27.9	25.3	20.6	18.2	(20)	
	2%	DB	23.6	28.7	36.3	43.1	46.9	46.3	42.1	38.5	37.4	35.8	31.4	26.0	(21)	
		MCWB	15.0	18.4	21.1	22.8	25.0	26.6	29.1	28.9	27.7	25.4	20.9	17.3	(22)	
	5%	DB	22.1	26.9	33.8	41.4	45.2	44.5	40.9	37.8	36.5	34.7	30.1	24.4	(23)	
		MCWB	14.4	17.6	21.0	22.8	24.8	26.5	29.1	28.9	27.6	24.7	20.0	16.3	(24)	
	10%	DB	20.8	25.1	31.8	39.3	43.6	43.0	39.7	37.0	35.6	33.7	28.7	23.0	(25)	
		MCWB	14.0	16.6	20.5	22.5	24.7	27.2	28.9	28.8	27.5	24.0	19.6	15.2	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	17.3	20.7	26.2	29.2	31.5	36.0	34.1	31.0	30.6	28.2	24.2	19.8	(27)	
		MCDB	22.9	27.7	33.9	39.7	43.2	42.9	38.4	36.6	36.3	34.0	30.1	25.9	(28)	
	2%	WB	16.1	19.6	23.5	26.4	29.5	31.5	31.1	30.1	29.6	27.4	22.7	18.4	(29)	
		MCDB	21.8	26.5	32.0	37.1	40.7	40.7	38.1	36.3	35.3	33.2	28.9	24.4	(30)	
	5%	WB	15.4	18.6	22.2	25.0	28.0	29.7	30.2	29.6	29.0	26.7	21.7	17.3	(31)	
		MCDB	20.8	25.5	30.8	36.0	39.4	39.4	37.7	35.9	34.5	32.5	27.7	22.8	(32)	
	10%	WB	14.6	17.5	21.2	24.0	26.9	28.9	29.7	29.2	28.4	26.0	20.7	16.3	(33)	
		MCDB	19.8	23.9	29.9	35.9	38.8	39.4	37.4	35.5	33.9	31.7	26.7	21.6	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	11.5	11.9	12.5	13.8	14.1	12.2	9.1	7.5	8.0	10.9	12.3	12.2	(35)	
		MCDBR	13.3	13.9	14.7	16.3	16.3	14.9	10.8	8.5	9.0	11.6	13.4	13.6	(36)	
	5% WB	MCWBR	6.7	6.6	5.5	5.6	5.3	4.7	3.4	3.0	3.2	4.2	5.3	6.8	(37)	
		MCWBR	11.9	12.4	13.2	14.3	14.7	12.0	9.4	7.9	8.0	9.6	11.3	12.1	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.487	0.502	0.535	0.595	0.627	0.714	0.814	0.733	0.594	0.542	0.568	0.512	(40)	
		taud	1.855	1.862	1.800	1.710	1.668	1.559	1.453	1.579	1.788	1.790	1.705	1.807	(41)	
	Ebn at Noon		722	757	762	732	710	646	583	627	705	709	637	671	(42)	
		Edh at Noon	175	188	213	240	250	278	308	270	212	199	203	177	(43)	
(44) All-Sky Solar Radiation	RadAvg		3.78	4.62	5.80	6.75	7.36	7.15	6.63	6.35	6.17	5.39	4.17	3.59	(44)	
	RadStd		0.22	0.34	0.28	0.18	0.15	0.20	0.32	0.37	0.26	0.19	0.19	0.17	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (4 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page